

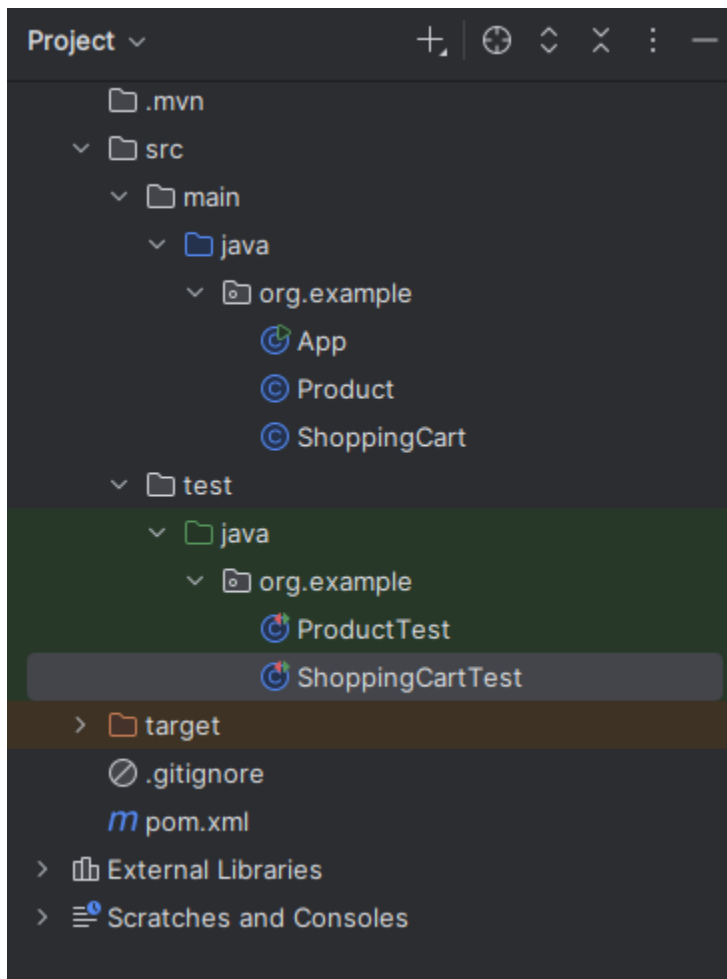
Capstone Project – Junit and Jmeter

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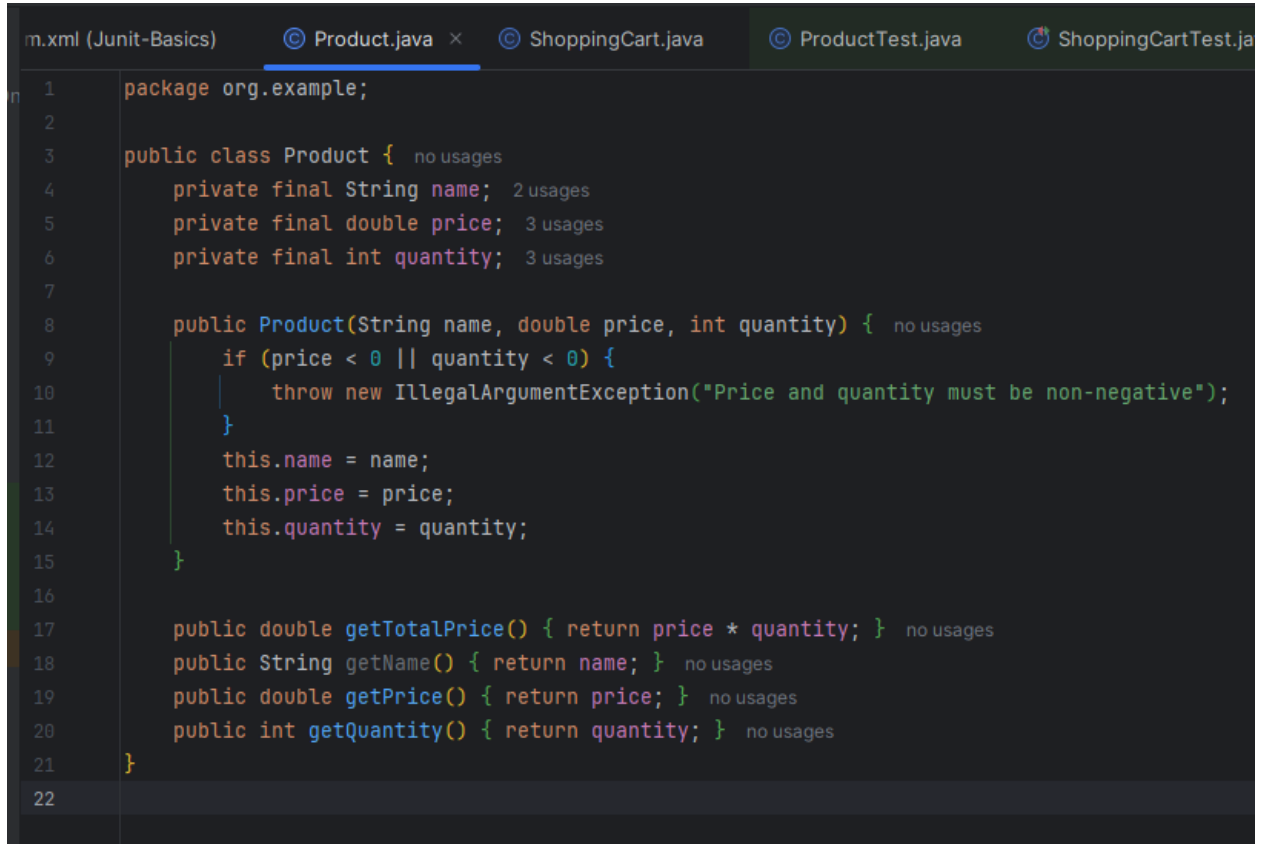
Question 1: JUnit (E-commerce application)

Sol :

Folder Structure:



Product.java file :



```
m.xml (JUnit-Basics)  Product.java  ShoppingCart.java  ProductTest.java  ShoppingCartTest.java
1  package org.example;
2
3  public class Product { no usages
4      private final String name; 2 usages
5      private final double price; 3 usages
6      private final int quantity; 3 usages
7
8      public Product(String name, double price, int quantity) { no usages
9          if (price < 0 || quantity < 0) {
10             throw new IllegalArgumentException("Price and quantity must be non-negative");
11         }
12         this.name = name;
13         this.price = price;
14         this.quantity = quantity;
15     }
16
17     public double getTotalPrice() { return price * quantity; } no usages
18     public String getName() { return name; } no usages
19     public double getPrice() { return price; } no usages
20     public int getQuantity() { return quantity; } no usages
21 }
22
```

ShoppingCart.java file :

```
xml (JUnit-Basics)  © Product.java  © ShoppingCart.java ×  © ProductTest.java  © ShoppingCartTest.java

1  package org.example;
2
3
4  import java.util.ArrayList;
5  import java.util.List;
6
7  public class ShoppingCart { 3 usages
8      private final List<Product> products = new ArrayList<>(); 4 usages
9
10     public void addProduct(Product product) { 5 usages
11         if (product == null) throw new IllegalArgumentException("Product cannot be null");
12         if (product.getPrice() == 0.0 && product.getQuantity() == 0) return;
13         products.add(product);
14     }
15
16     public double getTotalPrice() { 4 usages
17         double total = 0.0;
18         for (Product p : products) total += p.getTotalPrice();
19         return total;
20     }
21
22     public int getProductCount() { return products.size(); } 4 usages
23     public List<Product> getProducts() { return products; } 1 usage
24 }
25
```

ProductTest.java file :

```
import org.example.Product;
import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;

class ProductTest {

    @Test
    void testGetTotalPrice() {
        Product product = new Product(name: "laptop", price: 1000.0, quantity: 2);
        assertEquals(expected: 2000.0, product.getTotalPrice());
    }

    @Test
    void testNegativePriceThrowsException() {
        Exception exception = assertThrows(IllegalArgumentException.class,
            () -> new Product(name: "Iphone", price: -1000.0, quantity: 1));
        assertEquals(expected: "Price and quantity must be non-negative", exception.getMessage());
    }

    @Test
    void testNegativeQuantityThrowsException() {
        Exception exception = assertThrows(IllegalArgumentException.class,
            () -> new Product(name: "tablet", price: 500.0, quantity: -3));
        assertEquals(expected: "Price and quantity must be non-negative", exception.getMessage());
    }
}
```

ShoppingCartTest.java file:

```
package org.example;

import org.example.Product;
import org.example.ShoppingCart;
import org.junit.jupiter.api.BeforeEach;
import org.junit.jupiter.api.Test;

import static org.junit.jupiter.api.Assertions.*;

class ShoppingCartTest {

    private ShoppingCart cart; 15 usages

    @BeforeEach
    void setUp() {
        cart = new ShoppingCart();
    }

    @Test
    void testAddProductAndCount() {
        Product product = new Product( name: "Book", price: 100.0, quantity: 2);
        cart.addProduct(product);
        assertEquals( expected: 1, cart.getProductCount());
        assertTrue(cart.getProducts().contains(product));
    }
}
```

```

@Test
void testGetTotalPriceMultipleProducts() {
    Product p1 = new Product( name: "Pen", price: 10.0, quantity: 5);
    Product p2 = new Product( name: "Notebook", price: 50.0, quantity: 2);
    cart.addProduct(p1);
    cart.addProduct(p2);
    assertEquals( expected: 150.0, cart.getTotalPrice());
}

@Test
void testEmptyCart() {
    assertEquals( expected: 0, cart.getProductCount());
    assertEquals( expected: 0.0, cart.getTotalPrice());
}

@Test
void testZeroPriceAndQuantityProduct() {
    double beforeTotal = cart.getTotalPrice();
    int beforeCount = cart.getProductCount();

    cart.addProduct(new Product( name: "Freebie", price: 0.0, quantity: 0));

    assertEquals(beforeTotal, cart.getTotalPrice(), delta: 0.0001);
    assertEquals(beforeCount, cart.getProductCount());
}

```

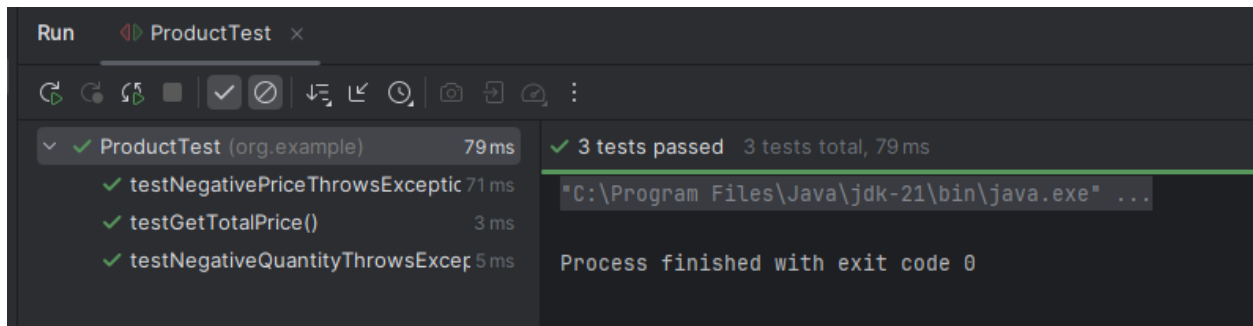
```

@Test
void testAddNullProductThrowsException() {
    Exception exception = assertThrows(IllegalArgumentException.class,
        () -> cart.addProduct(null));
    assertEquals( expected: "Product cannot be null", exception.getMessage());
}
}

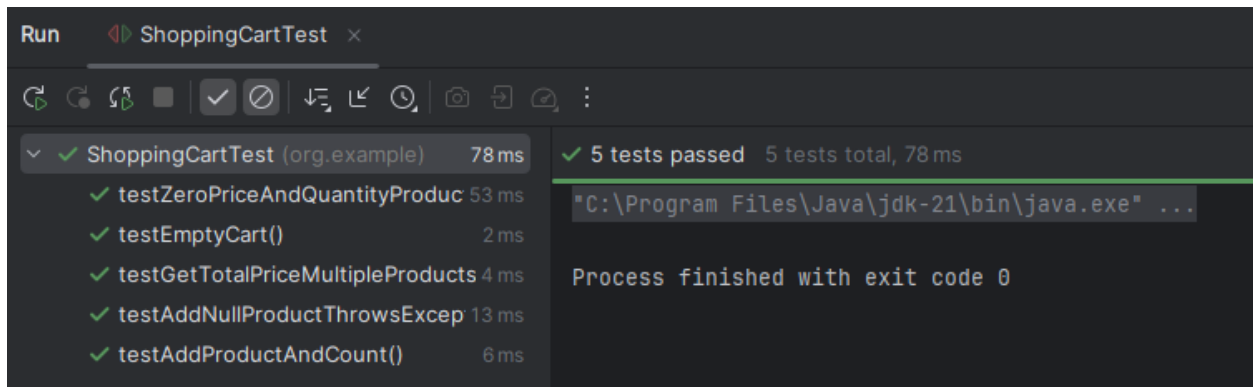
```

Results:

ProductTest :



ShoppingCartTest :



Question 2: Jmeter

Steps:

1. Open JMeter → File → New → create a new Test Plan → name it DevlopRestApiTestPlan.

2. Add Thread Group

Right-click Test Plan → Add → Threads (Users) → Thread Group
Set Number of Users = 20, Loop Count = 1.

3. Add HTTP Request Defaults

Right-click Thread Group → Add → Config Element → HTTP Request Defaults
Set Server Name = reqres.in, Protocol = https

4. Add HTTP Requests (Samplers)

GET: Get list of all objects → Path = /api/users?page=2

POST: Create object → Path = /api/users, Method = POST, Body = JSON

PUT: Update object → Path = /api/users/2, Method = PUT, Body = JSON

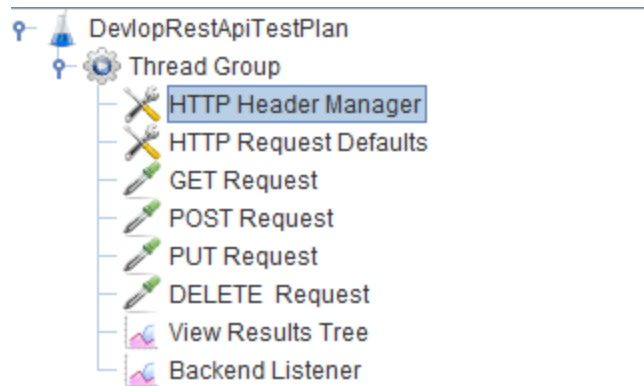
DELETE: Delete object → Path = /api/users/2, Method = DELETE

5. Add Listeners

Right-click Thread Group → Add → Listener → View Results Tree / Graph Results
/Backend Listener (for InfluxDB)

6. Save Test Plan → DevlopRestApiTestPlan.jmx.

Folder Structure:



Thread Group

Name:

Comments:

Action to be taken after a Sampler error

☒ Continue ☐ Start Next Thread Loop ☐ Stop Thread ☐ Stop Test ☐ Stop Test Now

Thread Properties

Number of Threads (users):

Ramp-up period (seconds):

Loop Count: ☐ Infinite

☒ Same user on each iteration

☐ Delay Thread creation until needed

☐ Specify Thread lifetime

Duration (seconds):

Startup delay (seconds):

```
graph TD
    A[DevlopRestApiTestPlan] --> B[Thread Group]
    B --> C[HTTP Header Manager]
    B --> D[HTTP Request Defaults]
    B --> E[GET Request]
    B --> F[POST Request]
    B --> G[PUT Request]
    B --> H[DELETE Request]
    B --> I[View Results Tree]
    B --> J[Backend Listener]
```

HTTP Header Manager

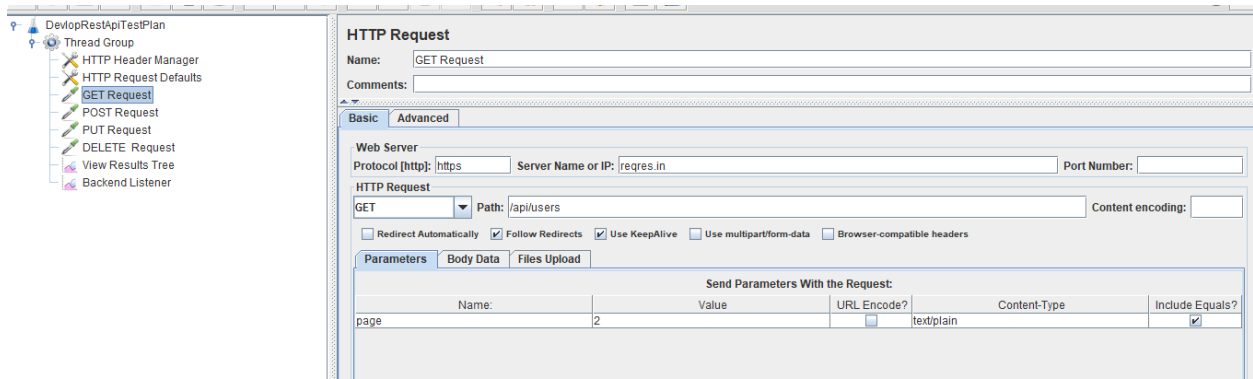
Name:

Comments:

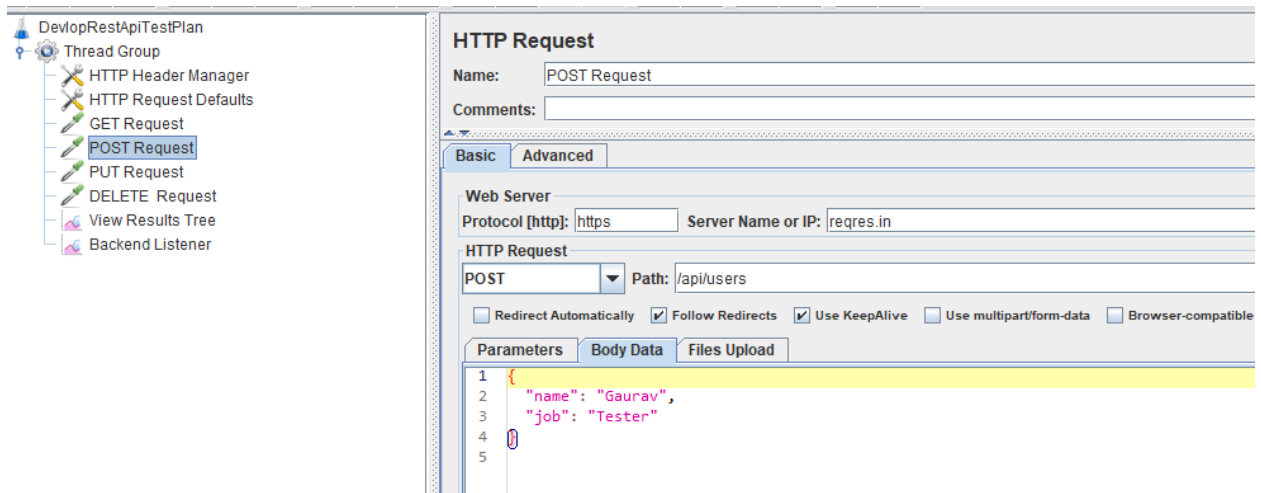
Headers Stored in the Header Manager

Name	Value
x-api-key	reqres-free-v1

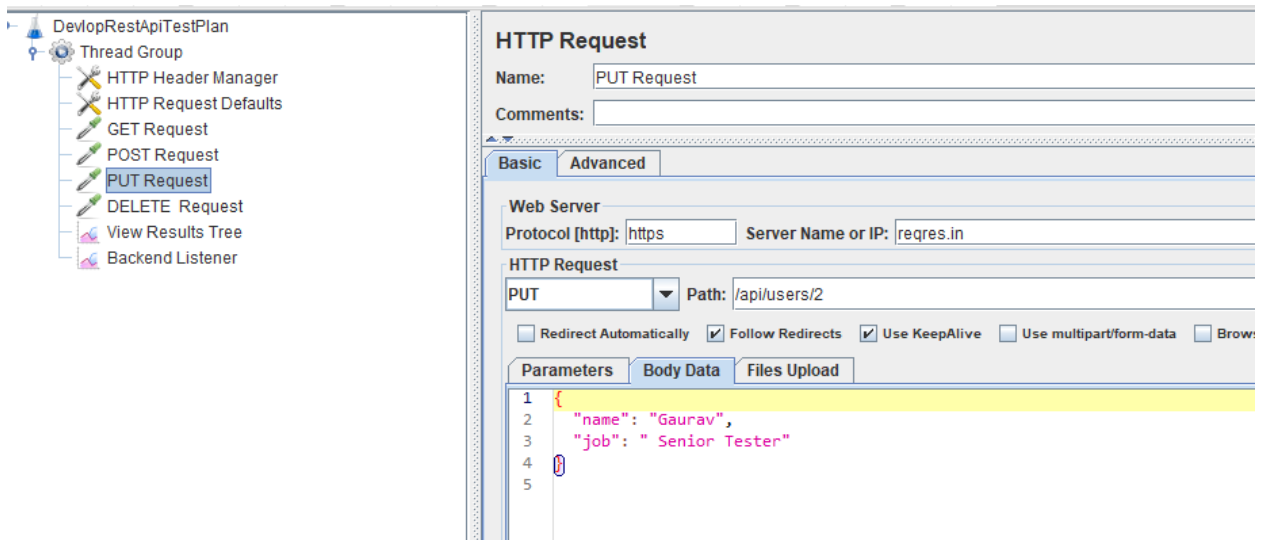
GET Request:



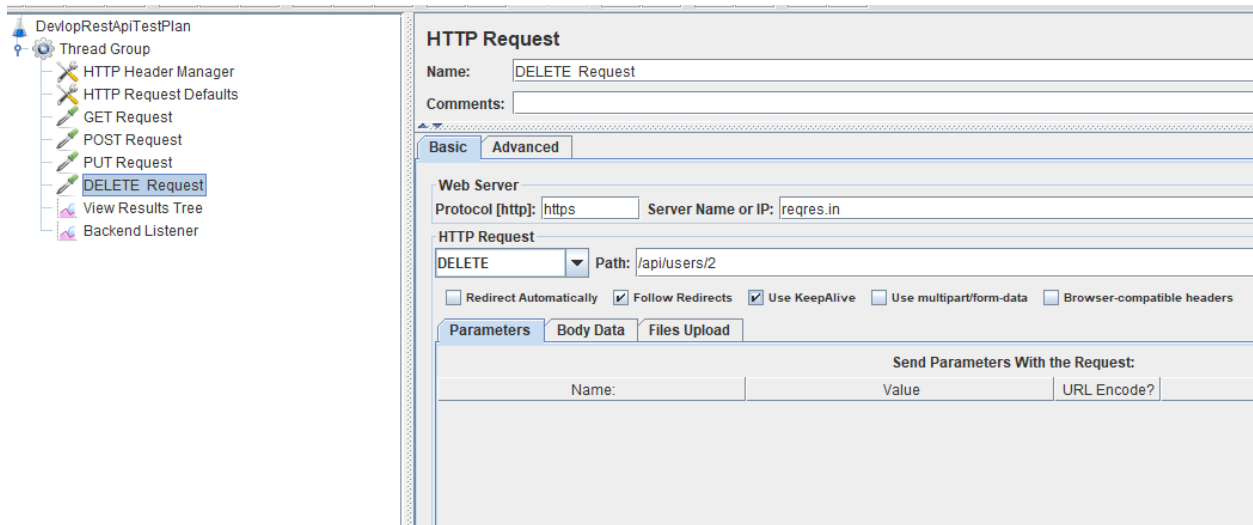
POST Request:



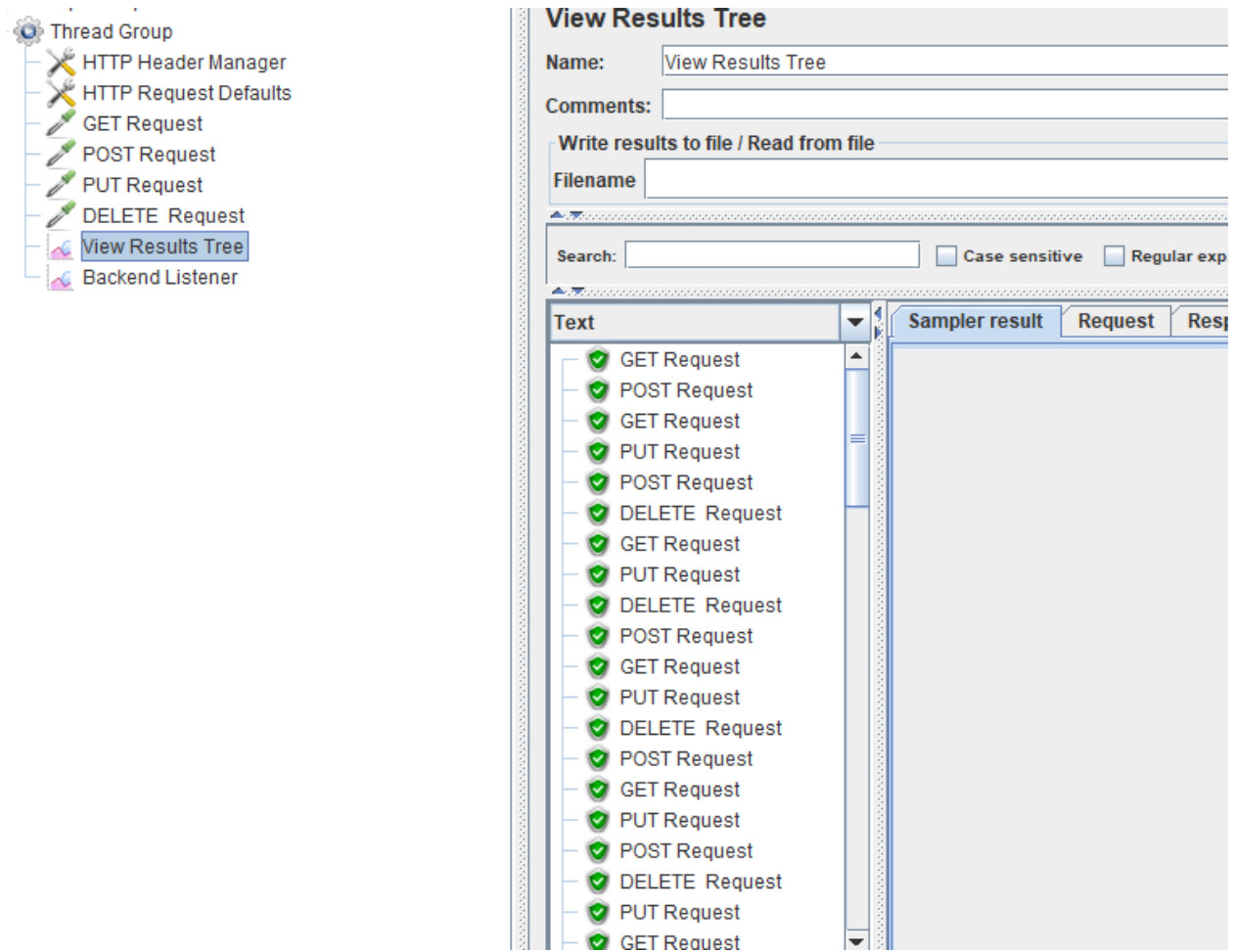
PUT Request:



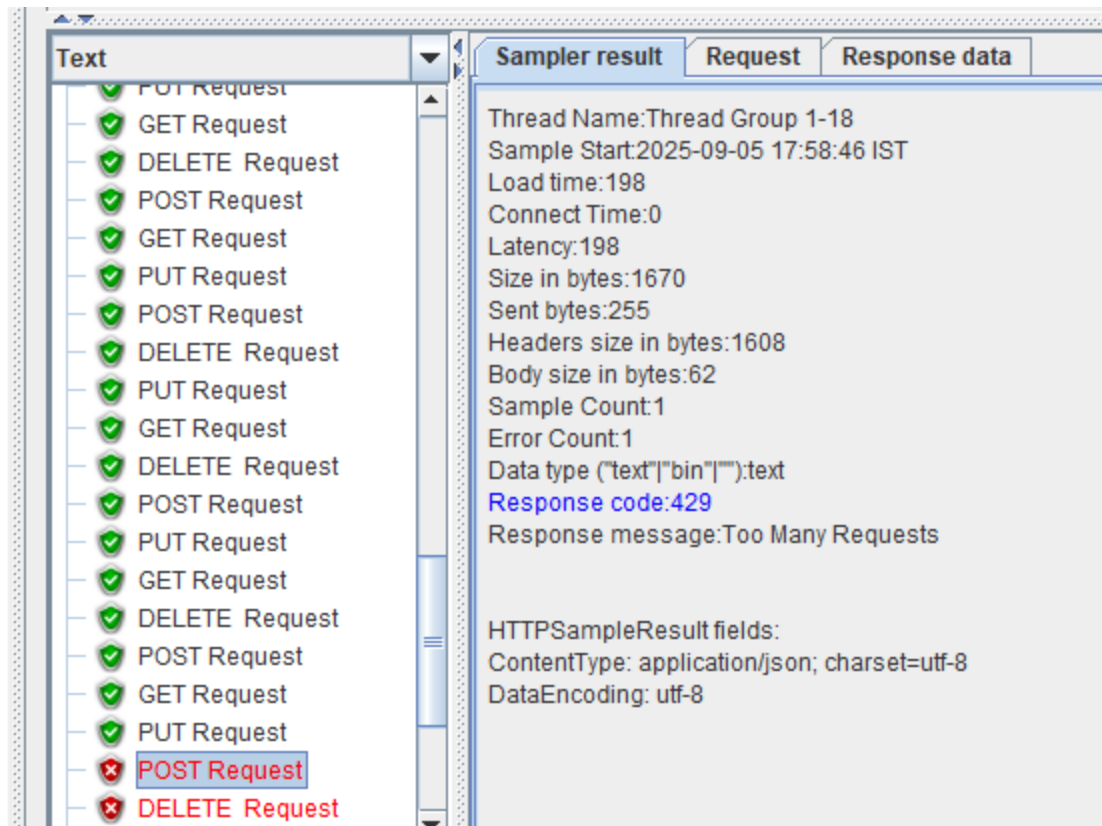
DELETE Request:



View Results Tree:



Few results Failed - > too many requests



2. Report Generation using Non GUI Mode steps:

1. Open JMeter bin folder in Command Prompt

```
cd "C:\Users\gaura\Downloads\apache-jmeter-5.6.3\apache-jmeter-5.6.3\bin"
```

2. Run the Test Plan in Non-GUI Mode

```
jmeter -n -t "C:\Users\gaura\OneDrive\Desktop\Capstone junit &
jmeter\DevlopRestApiTestPlan.jmx" -l "C:\Users\gaura\OneDrive\Desktop\Capstone
junit & jmeter\result.jtl"
```

-n → Non-GUI mode

-t → Path to .jmx test plan

-l → Log/result file path

3. Generate HTML Report from Test Results

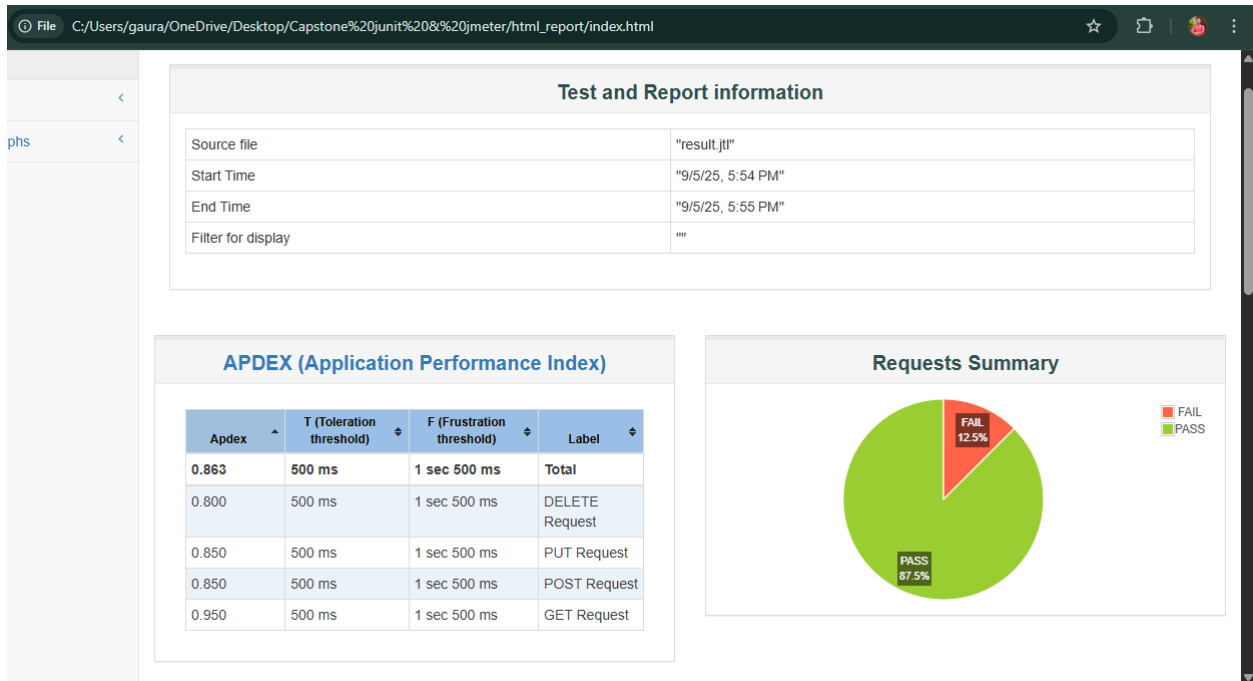
```
jmeter -g "C:\Users\gaura\OneDrive\Desktop\Capstone junit & jmeter\result.jtl" -o  
"C:\Users\gaura\OneDrive\Desktop\Capstone junit & jmeter\JMeterHTMLReport"
```

4. View HTML Report

Open the folder JMeterHTMLReport → open index.html in browser to see the report with graphs and metrics.

```
The directory name is invalid.  
C:\Users\gaura\Downloads\apache-jmeter-5.6.3\apache-jmeter-5.6.3\bin>jmeter -n -t "C:\Users\gaura\OneDrive\Desktop\Capstone junit & jmeter\DevlopRestApiTestPlan.jmx" -l "C:\Users\gaura\OneDrive\Desktop\Capstone junit & jmeter\result.jtl" -e -o "C:\Users\gaura\OneDrive\Desktop\Capstone junit & jmeter\html_report"  
WARN StatusConsoleListener The use of package scanning to locate plugins is deprecated and will be removed in a future release  
WARN StatusConsoleListener The use of package scanning to locate plugins is deprecated and will be removed in a future release  
WARN StatusConsoleListener The use of package scanning to locate plugins is deprecated and will be removed in a future release  
WARN StatusConsoleListener The use of package scanning to locate plugins is deprecated and will be removed in a future release  
Creating summariser <summary>  
Created the tree successfully using C:\Users\gaura\OneDrive\Desktop\Capstone junit & jmeter\DevlopRestApiTestPlan.jmx  
Starting standalone test @ 2025 Sep 5 17:54:51 IST (1757075091887)  
Waiting for possible Shutdown/StopTestNow/HeapDump/ThreadDump message on port 4445  
summary + 59 in 00:00:08 = 7.6/s Avg: 292 Min: 192 Max: 1057 Err: 0 (0.00%) Active: 2 Started: 16 Finished: 14  
summary + 21 in 00:00:03 = 7.7/s Avg: 226 Min: 192 Max: 326 Err: 10 (47.62%) Active: 0 Started: 20 Finished: 20  
summary = 80 in 00:00:11 = 7.6/s Avg: 275 Min: 192 Max: 1057 Err: 10 (12.50%)  
Tidying up ... @ 2025 Sep 5 17:55:02 IST (1757075102761)  
... end of run  
C:\Users\gaura\Downloads\apache-jmeter-5.6.3\apache-jmeter-5.6.3\bin>
```

```
C:\Users\gaura\Downloads\apache-jmeter-5.6.3\apache-jmeter-5.6.3\bin>jmeter -n -t "C:\Users\gaura\OneDrive\Desktop\Capstone junit & jmeter\DevlopRestApiTestPlan.jmx" -l "C:\Users\gaura\OneDrive\Desktop\Capstone junit & jmeter\result2.jtl"  
WARN StatusConsoleListener The use of package scanning to locate plugins is deprecated and will be removed in a future release  
WARN StatusConsoleListener The use of package scanning to locate plugins is deprecated and will be removed in a future release  
WARN StatusConsoleListener The use of package scanning to locate plugins is deprecated and will be removed in a future release  
WARN StatusConsoleListener The use of package scanning to locate plugins is deprecated and will be removed in a future release  
Creating summariser <summary>  
Created the tree successfully using C:\Users\gaura\OneDrive\Desktop\Capstone junit & jmeter\DevlopRestApiTestPlan.jmx  
Starting standalone test @ 2025 Sep 5 18:33:41 IST (1757077421060)  
Waiting for possible Shutdown/StopTestNow/HeapDump/ThreadDump message on port 4445  
summary = 80 in 00:00:11 = 7.5/s Avg: 267 Min: 194 Max: 472 Err: 60 (75.00%)  
Tidying up ... @ 2025 Sep 5 18:33:52 IST (1757077432363)  
... end of run  
C:\Users\gaura\Downloads\apache-jmeter-5.6.3\apache-jmeter-5.6.3\bin>
```



Statistics

Requests	Executions			Response Times (ms)							Throughput	Network (KB/sec)	
Label ^	#Samples ^	FAIL ^	Error % ^	Average ^	Min ^	Max ^	Median ^	90th pct ^	95th pct ^	99th pct ^	Transactions/s ^	Received ^	Sent ^
Total	80	10	12.50%	275.44	192	1057	233.50	381.90	432.10	1057.00	7.73	14.23	1.71
DELETE Request	20	4	20.00%	223.50	192	265	219.00	257.80	264.65	265.00	2.28	3.46	0.52
GET Request	20	0	0.00%	316.70	203	1057	248.00	653.50	1038.15	1057.00	2.09	5.53	0.31
POST Request	20	3	15.00%	295.30	197	433	320.00	393.70	431.10	433.00	2.26	3.63	0.56
PUT Request	20	3	15.00%	266.25	192	482	231.00	387.00	477.30	482.00	2.27	3.62	0.59

Errors

Type of error ^	Number of errors v	% in errors ^	% in all samples ^
429/Too Many Requests	10	100.00%	12.50%

Errors												
Type of error	Number of errors		% in errors		% in all samples							
429/Too Many Requests	10		100.00%		12.50%							

Top 5 Errors by sampler												
Sample	#Samples	#Errors	Error	#Errors	Error	#Errors	Error	#Errors	Error	#Errors	Error	#Errors
Total	80	10	429/Too Many Requests	10								
DELETE Request	20	4	429/Too Many Requests	4								
POST Request	20	3	429/Too Many Requests	3								
PUT Request	20	3	429/Too Many Requests	3								

3. Connect JMeter with influx db. and store data in Database DBName : DevInfluxDB

Steps:

1. make sure influxdb server is running and database DevInfluxDB is created
2. open your jmeter test plan DevlopRestApiTestPlan.jmx
3. right click thread group, add listener, select backend listener
4. in backend listener implementation, choose
org.apache.jmeter.visualizers.backend.influxdb.InfluxdbBackendListenerClient
5. set url as <http://127.0.0.1:8086/write?db=DevInfluxDB>, application name as DevlopRestApiTestPlan, measurement as jmeter, summary only as false

6. save the test plan
7. run the test in non-gui mode, metrics will be stored in DevInfluxDB

```
C:\Users\gaura\Downloads\influxdb-1.11.8_windows_amd64>influx.exe -host 127.0.0.1 -port 8086
Connected to http://127.0.0.1:8086 version v1.11.8
InfluxDB shell version: v1.11.8
> CREATE DATABASE DevInfluxDB;
> show database;
ERR: error parsing query: found DATABASE, expected CONTINUOUS, DATABASES, DIAGNOSTICS, FIELD, GRANTS, MEASUREMENT, MEASUREMENTS, QUERIES, RETENTION,
SERIES, SHARD, SHARDS, STATS, SUBSCRIPTIONS, TAG, USERS at line 1, char 6
Warning: It is possible this error is due to not setting a database.
Please set a database with the command "use <database>".
> SHOW DATABASES;
name: databases
name
----
_internal
DevInfluxDB
> USE DevInfluxDB;
Using database DevInfluxDB
> SHOW MEASUREMENTS;
name: measurements
name
----
events
jmeter
> SELECT * FROM jmeter LIMIT 5;
name: jmeter
time          application      avg      count countError endedT hit max maxAT meanAT min minAT pct90.0      pct95.0      p
ct99.0 rb      responseCode responseMessage sb      startedT  statut transaction
-----
1757077381610000000 application name          0          internal 0          0          0          0          0          0          0          0          0
1757077386608000000 application name 298.30555555555554 36 0          36 627          196          412.90000000000001 497.7999999999998 6
27 68743          8891          all          all          9          375          210          375          375          3
1757077386616000000 application name 265.8888888888889 9          PUT Request
75 14639          2376          all          PUT Request
```

Add backend listener in JMeter:

DevopRestApiTestPlan

- Thread Group
 - HTTP Header Manager
 - HTTP Request Defaults
 - GET Request
 - POST Request
 - PUT Request
 - DELETE Request
 - View Results Tree
 - Backend Listener

Backend Listener

Name: Backend Listener

Comments:

Backend Listener implementation: `org.apache.jmeter.visualizers.backend.influxdb.InfluxdbBackendListenerClient`

Async Queue size: 5000

Parameters	
Name	Value
influxdbMetricsSender	org.apache.jmeter.visualizers.backend.influxdb.HttpMetricsSender
influxdbUrl	http://127.0.0.1:8086/write?db=DevInfluxDB
application	application name
measurement	jmeter
summaryOnly	false
samplersRegex	*
percentiles	99,95,90
testTitle	Test name
eventTags	

4. Connect influxdb with Grafana and create a dashboard for showing the result of test plan.

Steps:

1. open grafana in browser at <http://localhost:3000> and login (admin/admin)
2. go to settings → data sources → add data source → choose influxdb
3. set url as <http://127.0.0.1:8086>, database as DevInfluxDB, version 1.x, leave user/password blank, then save and test
4. click + → dashboard → add new panel
5. select influxdb as data source and write query to fetch jmeter metrics
6. customize panel to show response time, requests per second, errors, etc.
7. add more panels for get, post, put, delete requests if needed
8. save the dashboard, metrics will update live when the test runs

```
17960055
9-05[18:37:45] Installing plugin logger=plugin.backgroundinstaller pluginId=grafana-lokiexplore-app version=
9-05[18:37:46] Updating plugin logger=plugin.installer pluginId=grafana-lokiexplore-app from=1.0.25 to=1.0.26
9-05[18:37:48] Failed to remove temporary filefileC:\Users\gaura\AppData\Local\Temp\2074832606.ziperrorremove C:\Users\gaura\AppData\Local\Te
832606.zip: The process cannot access the file because it is being used by another process. logger=plugin.repository
9-05[18:37:48] Downloaded and extracted grafana-lokiexplore-app v1.0.26 zip successfully to C:\Users\gaura\Downloads\grafana-enterprise_12.1.
967602_windows_amd64\grafana-12.1.1\data\plugins\grafana-lokiexplore-app logger=installer.fs
9-05[18:37:48] Plugin registered logger=plugins.registration pluginId=grafana-lokiexplore-app
9-05[18:37:48] Plugin successfully installed logger=plugin.backgroundinstaller pluginId=grafana-lokiexplore-app version= duration=
82s
9-05[18:37:48] Installing plugin logger=plugin.backgroundinstaller pluginId=grafana-pyroscope-app version=
9-05[18:37:49] Failed to remove temporary filefileC:\Users\gaura\AppData\Local\Temp\3856589991.ziperrorremove C:\Users\gaura\AppData\Local\Te
589991.zip: The process cannot access the file because it is being used by another process. logger=plugin.repository
9-05[18:37:49] Installing plugin logger=plugin.installer pluginId=grafana-pyroscope-app version=
9-05[18:37:50] Downloaded and extracted grafana-pyroscope-app v1.8.1 zip successfully to C:\Users\gaura\Downloads\grafana-enterprise_12.1.1_1
602_windows_amd64\grafana-12.1.1\data\plugins\grafana-pyroscope-app logger=installer.fs
9-05[18:37:50] Plugin registered logger=plugins.registration pluginId=grafana-pyroscope-app
9-05[18:37:50] Plugin successfully installed logger=plugin.backgroundinstaller pluginId=grafana-pyroscope-app version= duration=1.
s
9-05[18:38:08] Failed to authenticate request logger=authn.service client=auth.client.session error="user token not found"
9-05[18:38:08] Request Completed logger=context userId=0 orgId=0 uname= method=GET path=/ status=302 remote_addr=[::1]
size=7 duration=7.4172ms size=29 referer= handler=/ status_source=server
9-05[18:38:08] Failed to authenticate request logger=authn.service client=auth.client.session error="user token not found"
9-05[18:38:17] Failed to authenticate request logger=authn.service client=auth.client.session error="user token not found"
9-05[18:38:17] Request Completed logger=context userId=1 orgId=1 uname=admin method=GET path=/api/live/ws status=-1 re
dr=[::1] time_ms=4 duration=4.2663ms size=0 referer= handler=/api/live/ws status_source=server
9-05[18:38:20] Usage stats are ready to report logger=infra.usagstats
9-05[18:47:42] Completed cleanup jobs logger=cleanup duration=84.8291ms
9-05[18:47:46] Update check succeeded logger=plugins.update.checker duration=2.1027607s
9-05[18:57:42] Completed cleanup jobs logger=cleanup duration=73.0481ms
9-05[18:57:46] Update check succeeded logger=plugins.update.checker duration=2.1665398s
9-05[19:07:42] Completed cleanup jobs logger=cleanup duration=34.6989ms
9-05[19:07:46] Update check succeeded logger=plugins.update.checker duration=2.3123945s
9-05[19:08:20] Usage stats are ready to report logger=infra.usagstats
```

Home

Bookmarks

Starred

Dashboards

Explore

Drilldown

Alerting

Connections

Add new connection

Data sources

Administration

Home > Connections > Data sources > influxdb

Q Search... ctrl+k + v

Database Access

Setting the database for this datasource does not deny access to other databases. The InfluxDB query syntax allows switching the database in the query. For example: `SHOW MEASUREMENTS ON _internal` or `SELECT * FROM "_internal"."database" LIMIT 10`

To support data isolation and security, make sure appropriate permissions are configured in InfluxDB.

Database	DevInfluxDB	
User	admin	
Password	configured	Reset
HTTP Method	Choose	
Min time interval	10s	
Autocomplete range	12h	
Max series	1000	

Delete

Save & test

